

Title (Not “Coaching How To Search”)

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Abstract. The abstract should briefly summarize the contents of the paper in 150–250 words.

Keywords: First keyword · Second keyword · Another keyword.

1 Introduction

2 Related Literature

3 Background and Preliminaries

Utilizing learning semantics presented in previous work on Machine Coaching [4, 3], we empirically explore its particular application on search coaching, by having a machine learner iteratively seek advice by a machine coach, emulating a human coach, as in [2], with the aim of iteratively improving its search capacity, guided by the coach’s explicit search policy. Given a fixed goal state, g , the learner applies its search policy, P_{learner} , to identify a path, p , starting from an arbitrary state, s , and ending at a goal state, g . Upon lack of knowledge, the learner seeks advice from its coach, which responds according to its own policy, P_{coach} , providing a single piece of advice to the learner. Each piece of advice has the form of an “if–then” rule, where the rule’s *body* (“if” part) is state–dependent and its *head* (the “then” part) is always an action to be undertaken by the learner. Rules are prioritized by the coach as in previous works [2, 3, 1] with later received ones by default overriding previous ones in P_{learner} , besides cases where the coach demands otherwise. In this setting, a *coaching step* is a single learner request for advice and the corresponding coach response, while a *coaching session* comprises of the full learning cycle, ending when the learner eventually crafts a path as requested (cf Algorithm 1). In all cases the learner seeks advice upon lack of knowledge for further searching, returning a partial path p starting from s .

We have considered the particular case of sorting formulated as a search problem, implementing two coaching *policies*, P_{bubble} , which relies on bubble sort, and P_{quick} , which relies on quicksort, both resulting to advice corresponding to the next step of the algorithm described by each coaching policy. A state

Data: $s, g, learner, coach$

Result: $learner$ with updated search policy; $path$ from s to g .

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do
  |  $path \leftarrow learner.search\_path(s, g);$ 
  |  $advice \leftarrow coach.evaluate(path, s, g);$ 
  |  $learner.update\_hypothesis(advice);$ 
while  $advice$  is not null;

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Algorithm 1: Search coaching process. s, g denote the start and goal states, respectively.

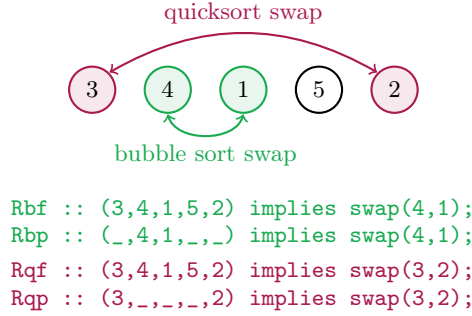


Fig. 1: Examples of full (f) and partial (p) advice from P_{bubble} (b) and P_{quick} (q) on state $(3, 4, 1, 5, 2)$.

of size n in this case is any permutation of $\{1, 2, \dots, n\}$, with start states being arbitrarily chosen among all possible states of the same size, while the goal state is always $(1, 2, \dots, n)$. Rules in this case have a potentially partial permutation of $\{1, 2, \dots, n\}$ as their body, and a swap action between two items of this permutation as their head, as shown in Figure 1.

In this setting, we define two types of advice: (A1) *full advice*, where each piece of advice depends on the entire state, and; (A2) *partial advice*, where each piece of advice depends on the specific part of the state to be improved. Moreover, for each advice type we have considered three learner configurations regarding advice memorization by the learner: (M1) *no memory* across different iterations for the same state size, n , i.e., coaching sessions and the corresponding learned policies are pairwise independent; (M2) *short memory* across different iterations for the same state size, n , i.e., coaching sessions across different values of n are independent but not within the same value of n , and; (M3) *long memory* across all values of state size, n , i.e., all coaching sessions are correlated, incrementally building on knowledge from all previous test cases. A coach might adopt one of the following *coaching styles*, as illustrated in Figure 2: (C1) *reactive* coaching, where the coach provides advice on the learner's last wrong decision in p ; (C2) *reflexive* coaching, where the coach provides advice at a random wrong decision in p , and; (C3) *proactive* coaching, where the coach provides advice on the learner's earliest wrong decision in p .

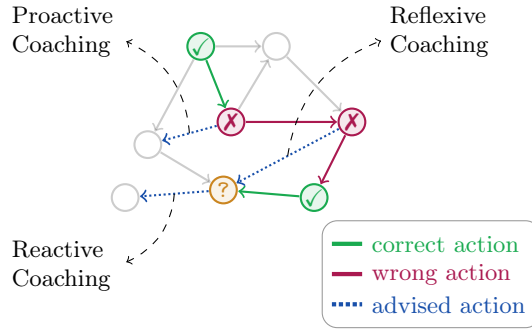


Fig. 2: Different coaching styles used in our experimental setup. Upon the learner reaching a state from where there is no knowledge on how to proceed (?), a reactive piece of advice resolves this lack of knowledge, a proactive piece of advice addresses the first wrong decision in the learner’s search path, while a reflexive piece of advice addresses a random mistake / lack of knowledge of the learner.

4 Experimental Setup

We ran a series of experiments first to verify learning efficacy, where we let state size, n , range from 1 to 20, running $m = 100$ iterations for each value of n . We have consider all possible combinations of advice type (full and partial), memory configurations (none, short, long), coaching styles (reactive, reflexive, proactive), and coaching policies (P_{bubble} and P_{quick}), resulting into 36 different coaching settings.

In order to explore how much the learner conforms to the coach’s instructions in the various settings explored, we define *coaching conformity* to be the relative frequency of learner decisions conforming to coach decisions on the same states over all such decisions in a session. First, we sample $m = 20$ repetitions for state size $n \in \{5, 10, 15, 20\}$ and compute coaching conformity for each one, across all coach policies, coaching styles, and memory configurations. Then, aiming to shed more light on the parameters affecting coaching conformity, we introduce a new type of advice, *partial@k*, $2 \leq k \leq n$, where $k - 2$ randomly selected state variables are included into the advice condition, alongside the two required to describe the swap action of interest. Such advice captures intermediate levels of specificity between the two extremes explored so far, with *partial@2* corresponding to partial and *partial@n* to full advice for state size n . We also define *advice coverage* as the percentage of state variables that is included in the condition of a piece of advice. So, e.g., a rule like $R :: (_, 2, 1, _, 5) \text{ implies } \text{swap}(2, 5)$ has an advice coverage of $3/5 = 60\%$. Again, we ran $m = 20$ iterations for state size $n \in \{5, 10, 15, 20\}$, $2 \leq k \leq n$, for all coaching policies, coaching styles, and memory configurations.

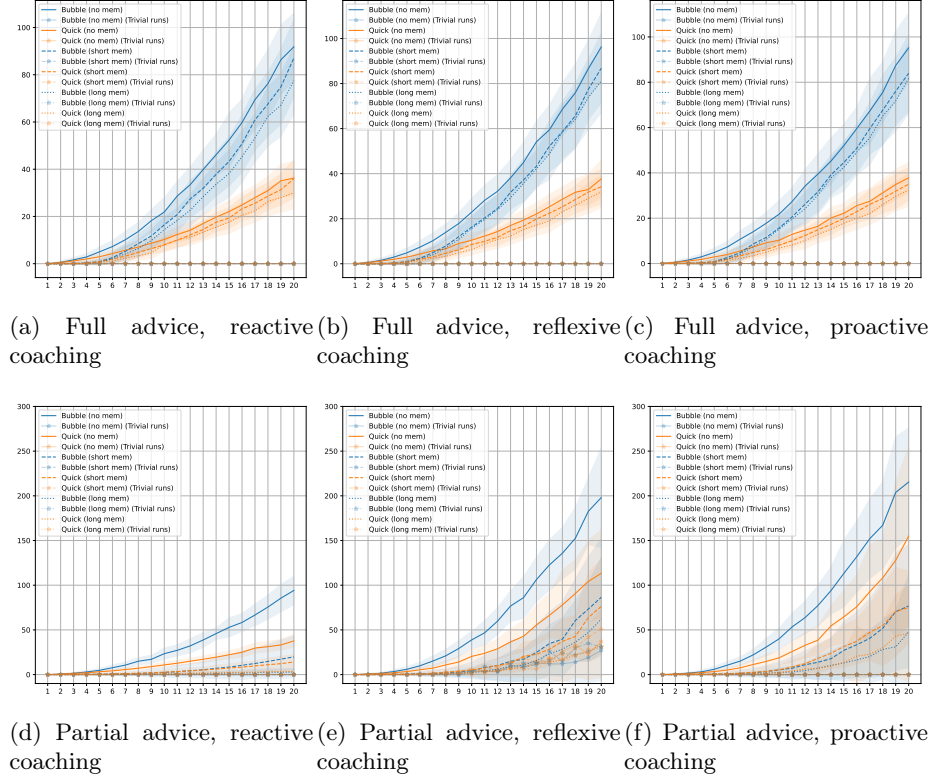


Fig. 3: Coaching steps as a function of state size, n . Full and partial advice (top and bottom, respectively) with reactive, reflexive and proactive coaching styles are considered (left, middle, and right, respectively), with all possible coach (P_{bubble} , P_{quick}) and memory (none, short, long) combinations are shown.

5 Results

In Figures 3 we present the number of coaching steps required to reach the goal state in each case as a function of the state size, n , for all three coaching styles. In all cases, except for partial advice reflexive coaching, the learner completed successfully all coaching sessions with the intended results. Considering reactive coaching and full advice (Fig. 3a), the utilization of either short or long memory does not appear to have any particular effect on the number of coaching steps, which was expected given the restricted applicability of provided advice. On the contrary, partial advice (Fig. 3d) combined with both short and long memory results to fewer coaching steps, which can be symmetrically explained by the broader applicability of the provided advice. Also observe that in the case of full advice, coaching steps are proportional to the big- O complexity of the underlying policy / sorting algorithm, which, again, comes as a result of limited advice

applicability, forcing the learner to essentially ask for step-by-step guidance at each separate case. In juxtaposition to the full advice case, utilizing memory significantly reduces the required coaching steps per session, since in this case the learner can effectively apply previously acquired knowledge to new cases. Furthermore, note that in the long memory case (Fig. 3d), coaching steps for both P_{bubble} and P_{quick} are essentially the same (dotted lines), indicating that in the case of reactive coaching, memorizing all previous advice eventually allows the learner to generalize and autonomously reach the goal state without frequent interaction with its “lazy” supervisor.

Considering proactive coaching, where the learner receives advice on its first wrong decision (Figs. 3c, 3f), we observe that full advice imposes the same trends in coaching step growth as with reactive coaching. However, providing partial advice leads to a significantly reduced number of coaching steps per session, as with reactive coaching, since, by utilizing memory, the learner is able to generalize from previous cases. Notably, coaching steps are systematically higher from the corresponding reactive coaching cases (Figs. 3d, 3f). This reflects the more “meticulous” way a proactive coach interacts with the learner, providing (even partial) advice that is more accurate, preventing any unnecessary exploration of the search space, at the cost of interacting more frequently with the learner, compared to a reactive coach, which just reacts to a learner’s demand for further knowledge.

Regarding reflexive coaching (Figs. 3b, 3e), the same trends as in proactive coaching are observed. However, in this case, the coach randomly selecting a state to intervene gives rise to the possibility that the same piece of advice might be provided more than once. We report the number of such sessions as *trivial runs*. As state size grows larger, so does the number of trivial runs, especially in non trivial memory configuration cases, where the learner’s policy is typically larger, hence it is, in general, more probable to receive the same piece of advice more than once.

Finally, we observe that coaching policy is irrelevant in case the learner is equipped with some kind of memory, regardless of advice type. For full advice this is expected given the restricted advice applicability which prohibits the learner from reusing previous knowledge, while partial advice allows the learner to generalize conforming to P_{coach} , reducing the need for coach feedback. Focusing on partial advice in the absence of memory, reactive coaching results to coaching sessions of size proportional to the coaching policy complexity, as noted above. However, for proactive and reflexive coaching this is not true, as the coach appears to interact with the learner more frequently, indicating that in the reactive case the learner might learn an efficacious but unfaithful policy, which might not occur for other coaching styles.

In Figure 4 we show average coaching conformity as a function of state size, n . Regardless of memory configuration, full advice combined with reactive coaching result to a virtually fully conformant learner, replicating the coach’s policy. Similar results are also obtained for proactive and reactive coaches. However, with partial advice, coaching conformity depends on the coaching style and the corre-

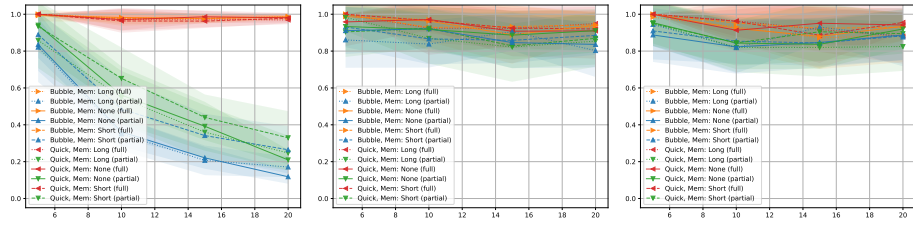


Fig. 4: Coaching fidelity against state size for reactive (left), reflexive (middle) and proactive (right) coaching.

sponding memory configuration. In particular, in the case of reactive coaching, partial advice results to virtually vanishing conformity for all memory configurations. For proactive coaching, however, partial advice results to a constantly high conformity for all examined state sizes, also barely affected by the corresponding memory configuration. Similarly, reflexive coaching also results to higher conformity than reactive coaching, roughly replicating results for proactive coaching. Also, observe how in all cases coach type does not affect conformity, except for partial advice reactive coaching, where any differences appear to vanish as state size grows. Overall, Figure 4 hints towards a trade-off between advice specificity

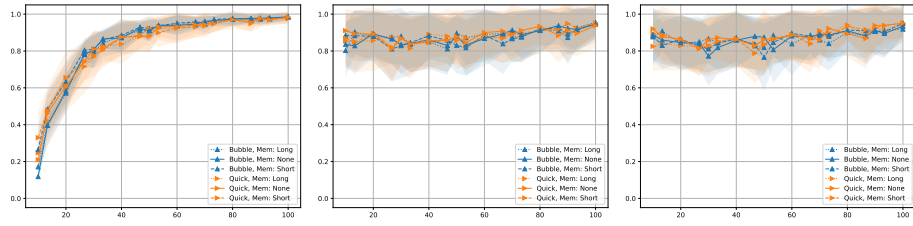


Fig. 5: Coaching fidelity against advice coverage for reactive (left), reflexive (middle) and proactive (right) coaching.

and coaching conformity. The more specific the advice the more conformant the learner to the coaching policy and, conversely, the looser the advice, the more the learner generalizes utilizing previous knowledge.

In Figure 5 we present coaching conformity against advice coverage for all coaching policies and coaching styles, aggregating over state size, n . Observe, at first, how coaching policy and memory configuration do not affect coaching conformity, while the most significant effect comes from different coaching styles. Indeed, in the case of reactive coaching, lower advice coverage results to the learner eventually reaching the goal state but in a way not conforming to the coach’s search policy, while the more specific the provided advice, the more accurate the learner in its searching steps towards the goal state. On the contrary, for

proactive and reflexive coaching, coaching conformity is just slightly increasing with respect to advice coverage, being relatively high, even for lower values of advice coverage. Compared to reactive coaching, proactive and reflexive coaching appear to guarantee a constant high conformity, mitigating the negative effects of arbitrarily low coverage almost equally well. In a sense, earlier coach intervention guarantees that the learner cannot deviate much from the coach’s guidance, even in case where advice is rather minimal, leaving space for generalization.

6 Conclusions

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